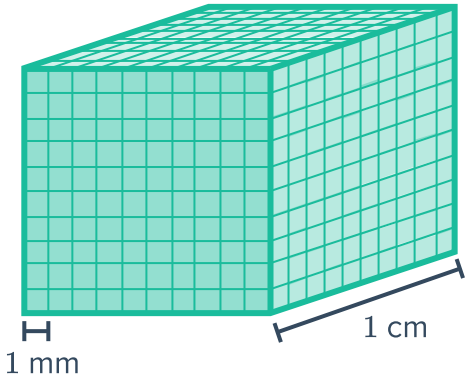
Units of volume and capacity

1 The cube shown in the diagram has a volume of $1\ \text{cm}^3$. Find its volume in cubic millimetres.



2 Convert the following to cubic millimetres:

a $3.5\ \text{cm}^3$ **b** $4.2\ \text{m}^3$ **c** $4\ \text{m}^3$ **d** $19\ \text{cm}^3$

3 Convert the following to cubic centimetres:

a $47\,600\ \text{mm}^3$ **b** $8.12\ \text{m}^3$ **c** $7000\ \text{mm}^3$ **d** $33\ \text{m}^3$

e $1\ \text{mL}$ **f** $8000\ \text{mL}$ **g** $40\ \text{mL}$ **h** $4.47\ \text{mL}$

4 What operation is required to convert cubic centimetres to cubic metres?

5 Convert the following to cubic metres:

a $1\,070\,000\,000\ \text{mm}^3$ **b** $51\,980\,000\ \text{cm}^3$ **c** $460\,000\,000\ \text{mm}^3$ **d** $16\,000\,000\ \text{cm}^3$

e $6000\ \text{L}$ **f** $800\ \text{L}$ **g** $18\ \text{kL}$ **h** $0.95\ \text{kL}$

6 Which metric unit represents 'one thousandth of a litre'?

7 Which metric unit represents 'one thousand millilitres'?

8 Convert the following to millilitres:

a $6\ \text{L}$ **b** $1.54\ \text{L}$ **c** $2\ \text{kL}$ **d** $0.17\ \text{kL}$

e $1\ \text{cm}^3$ **f** $8.5\ \text{cm}^3$ **g** $9000\ \text{mm}^3$ **h** $0.005\ \text{m}^3$

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9 Convert the following to litres:

a $4920\ \text{mL}$	**b** $5000\ \text{mL}$	**c** $9\ \text{kL}$	**d** $1.67\ \text{kL}$
e $18.62\ \text{m}^3$	**f** $2\ \text{m}^3$	**g** $6750\ \text{cm}^3$	**h** $800\ \text{cm}^3$

10 Convert the following to kilolitres:

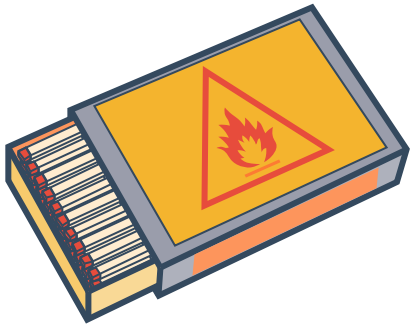
a $6570\ \text{L}$	**b** $84.6\ \text{L}$	**c** $1\,500\,000\ \text{mL}$	**d** $60\,000\ \text{mL}$
e $17\ \text{m}^3$	**f** $0.37\ \text{m}^3$	**g** $25\,000\,000\ \text{cm}^3$	**h** $54\,000\ \text{cm}^3$

Volume estimation

11 Consider the matchbox shown:

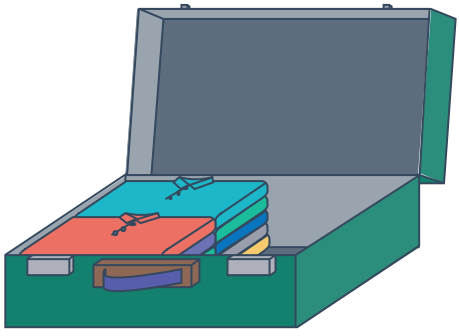
a State the most appropriate unit for measuring the volume of a match box.

b Estimate the volume of the matchbox.



12 The volume of the suitcase shown has a total capacity of $88\,500\ \text{cm}^3$.

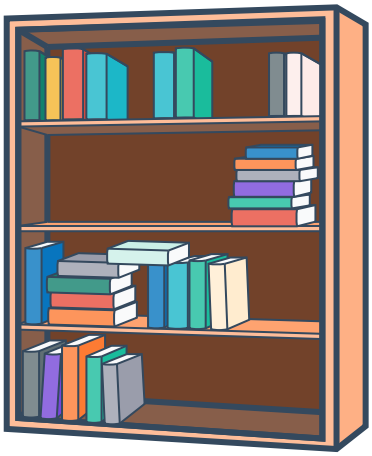
Estimate the volume of the contained shirts.



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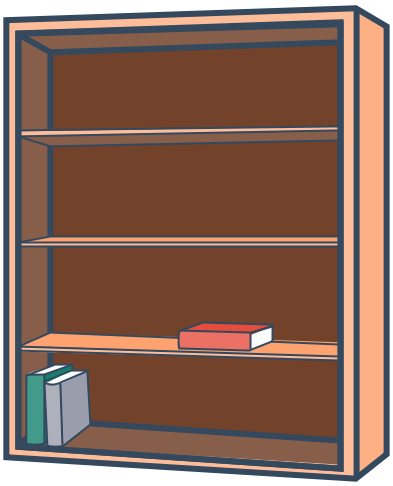
13 The volume of the bookshelf shown below is known to be $0.64\ \text{m}^3$.

Estimate the volume of the books on the bookshelf.



14 The volume of the bookshelf shown is known to be $0.68\ \text{m}^3$.

Estimate the volume of the books on the bookshelf.



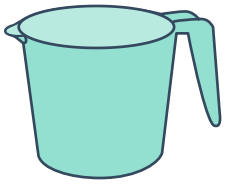
15 State the most appropriate unit for measuring the volume of the following:

a Sand pit **b** Wardrobe **c** Water glass **d** Swimming pool

e Air in a hall **f** Water bottle **g** Air in a balloon

Capacity estimation

16 Which units can we measure with a jug like the one in the picture?

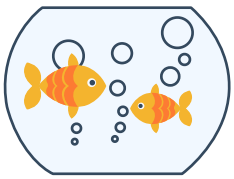


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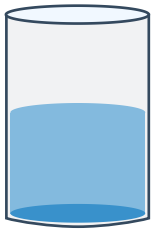
17 State the most appropriate unit for measuring the capacity of a cup of coffee.



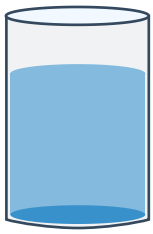
18 State the most appropriate unit for measuring the capacity of a fish bowl.



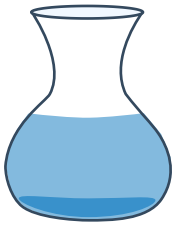
19 Estimate the volume of liquid in this container if it is able to hold $2400\ \text{mL}$ when full.



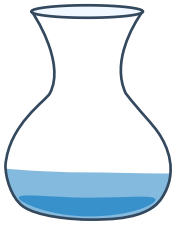
20 Estimate the volume of liquid in this container if it is able to hold $3.6\ \text{L}$ when full.



21 Estimate the volume of liquid in this container if it is able to hold $3600\ \text{mL}$ when full.



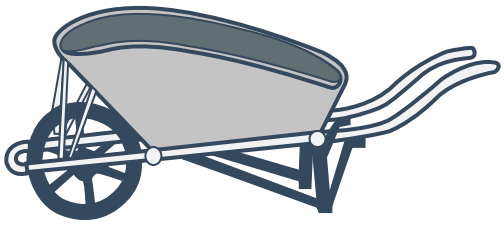
22 Estimate the volume of liquid in this container if it is able to hold 3.8 litres when full.



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23 A landscaper needs to move soil across the garden. Choose the best estimate for the capacity of soil that can be carried by a wheelbarrow in one trip.

A $771\ \text{mL}$ **B** $21\ \text{ML}$ **C** $71\ \text{L}$



24 State the most appropriate unit of measure for finding the volume of the following:

a Water in a dam.

b A dose of vaccine injection.

c A small carton of milk.

d The Atlantic Ocean.

e A dosage of medicine.

f A bottle of washing detergent.

Applications

25 Quiana bought a $400\ \text{mL}$ bottle of soft drink. How many litres of soft drink did she buy?

26 The following baby bottle holds $100\ \text{mL}$ of milk:

Find its volume in cm^3.



27 The following water bottle has a volume of $700\ \text{cm}^3$:

a Find its capacity in millilitres.

b Find its capacity in litres.



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28 This juice box contains $1600\ \text{mL}$ of juice:



a Find its volume in cm^3.

b Calculate the length of the juice box if it is $8\ \text{cm}$ tall and $5\ \text{cm}$ wide.

29 A cubic box has capacity of $343\ \text{L}$. Find its volume in cubic centimetres.

30 A small pond contains $3900\ \text{L}$ of water. Find the capacity of the pond in cubic metres.

31 A jug has a volume of $12\,395\ \text{cm}^3$. How many litres of soda water can it hold? Round your answer to three decimal places.

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